

EMODIN EXTRACTION METHODS FROM *POLYGONACEAE* PLANTS

METODE DE EXTRAȚIE A EMODINEI DIN PLANTE CE APARTIN FAMILIEI *POLYGONACEAE*

Lavinia-Diana-Nicoleta BARBU ^{1,2)}, Oana-Alina BOIU-SICUIA^{*1,2)}, Sorina DINU ¹⁾,
Cristina Maria LUMÂNARE ^{1,2)}, Daniel-Nicolae COJANU^{1,2)}, Narcisa BĂBEANU ²⁾

¹⁾ Research - Development Institute for Plant Protection, Bucharest / Romania;

²⁾ University of Agronomic Sciences and Veterinary Medicine of Bucharest, Faculty of Biotechnologies / Romania

*Tel: 004-021-2693231; *E-mail: sicuia_oana@yahoo.com

Keywords: anthraquinones, emodin, extraction methods, *Polygonaceae*, *Rumex*

ABSTRACT

Plants of the *Polygonaceae* family are mostly spontaneous herbs, widespread throughout the world. Although culinary they are less used, their medicinal properties make them highly appreciated. Emodin is one of the biological active compounds of pharmacological interest that can be extracted from these plants. This anthraquinone has many clinically proven medicinal properties. Its selective cytotoxic effect is now triggering the attention due to the treatment benefits in certain cancer types. In the present study, attention is drawn to emodin extraction methods and solvents used. Available information on biologically active compounds found in *Polygonaceae* plants, emodin synthesis pathways, factors modulating emodin levels in plants, as well as some pharmacological properties of this compound are also presented.

REZUMAT

Plantele din familia *Polygonaceae* se găsesc spontan și sunt răspândite în întreaga lume. Deși sunt mai puțin utilizate culinar, acestea sunt apreciate pentru proprietățile lor medicinale. Emodina este unul dintre compușii biologic activi de interes farmacologic ce poate fi extrasă din aceste plante. Aceasta este o antrachinonă cu numeroase proprietăți medicinale demonstrate clinic. În prezent, atenția fiind îndreptată către efectul citotoxic selectiv ce prezintă beneficii în tratarea anumitor forme de cancer. În această lucrare, atenția este îndreptată către metodele de extracție și solvenții utilizați pentru extracția emodinei. De asemenea, sunt prezentate informații disponibile privind compușii biologic activi prezenți în plantele din familia *Polygonaceae*, sinteza emodinei în plante, factorii stimulatori și proprietățile farmaceutice ale acestui compus.

INTRODUCTION

Emodin is an aromatic polyketide (figure 1) belonging to the anthraquinone family (Izhaki, 2002), a large group of natural compounds that can be found in certain plants (Akkol et al., 2021), fungi (Lu et al., 2010; Qui et al., 2021), and insects (Kikuchi et al., 2011). This compound can be found under various name, such as, casanthranol, CAS 518-82-1, E7881, emodol, schuttgelb, 1,3,8-trihydroxy-6-methyl-anthraquinone, 1,3,8-trihydroxy-6-methyl-anthracene-9,10-dione, 3-methyl-1,8-trihydroxyanthraquinone, but commonly it is referred as emodin (figure 1).

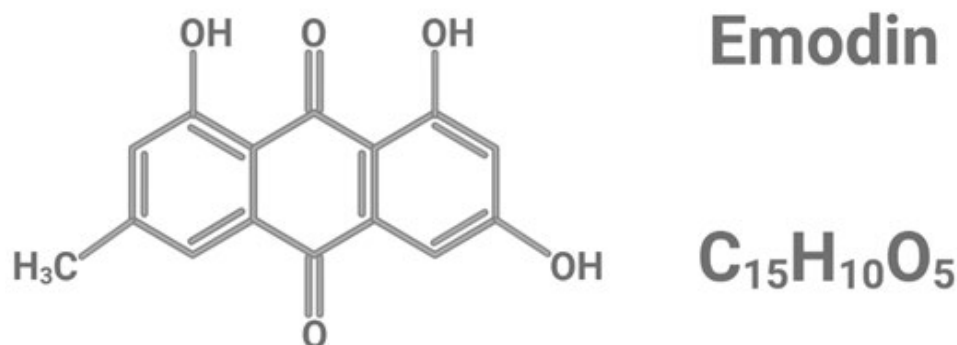


Fig. 1 - Emodin – chemical structure and molecular formula

Plants of the *Polygonaceae* family, such as *Rheum*, *Rumex* and *Polygonum* spp. are mentioned to be particularly rich in emodin. As well, some members of *Rubiaceae*, *Rhamnaceae*, *Fabaceae* and *Liliaceae* families are also referred as rich in this anthraquinone (Akkol et al., 2021). However, emodin has been found in at least 22 plant families (Table 1).

Table 1

Families, genera and part of plants containing emodin

Family	Genera	Plant parts	References
<i>Actinidiaceae</i>	<i>Actinidia</i>	root	Izhaki (2002)*
<i>Adoxaceae</i>	<i>Sambucus</i>	stem and root bark	Xiao et al. (2016)
<i>Amaranthaceae</i>	<i>Achyranthes</i>	root	Izhaki (2002)*
	<i>Amaranthus</i>	leaves	Ghosh et al. (2014)
<i>Asteraceae</i>	<i>Artemisia</i>	whole plant	Izhaki (2002)*
	<i>Lactuca</i>	whole plant	Izhaki (2002)*
	<i>Petasites</i>	rhizome	Izhaki (2002)*
<i>Bignoniaceae</i>	<i>Catalpa</i>	stem	Izhaki (2002)*
<i>Brassicaceae</i>	<i>Isatis</i>	root	Lin et al. (2005)
<i>Clusiaceae</i>	<i>Hypericum</i>	whole plant	Izhaki (2002)*
	<i>Ploiarium</i>	branch	Izhaki (2002)*; Ee and Ng (2004)
	<i>Psorospermum</i>	root, bark	Izhaki (2002)*
<i>Cupressaceae</i>	<i>Juniperus</i>	heartwood	Izhaki (2002)*
<i>Fabaceae</i>	<i>Cassia</i> (sin. <i>Senna</i>)	root, bark, stem, leave, heartwood, seed, pod	Izhaki (2002)*; Narayanan and Jadhav (2015a); Yue et al. (2018)
	<i>Phaseolus</i>	whole plant	Izhaki (2002)*
	<i>Pisum</i>	whole plant	Izhaki (2002)*
<i>Hypericaceae</i>	<i>Cratoxylum</i>	bark	Boonnak et al. (2006)
<i>Liliaceae</i>	<i>Aloe</i>	whole plant	Izhaki (2002)*; Narayanan and Jadhav (2015a)
<i>Myrsinaceae</i>	<i>Myrsine</i>	root	Izhaki (2002)*
<i>Orobanchaceae</i>	<i>Odontites</i>	whole plant	Zhang et al. (2021)
<i>Plantaginaceae</i>	<i>Plantago</i>	whole plant	Izhaki (2002)*
<i>Poaceae</i>	<i>Agropyron</i>	rhizome, root	Izhaki (2002)*
<i>Polygonaceae</i>	<i>Coccoloba</i>	root	de Barros et al. (2011)
	<i>Fallopia</i>	rhizome	Rehorska et al. (2013)
	<i>Polygonum</i>	rhizome, root, stem	Izhaki (2002)*; Lin et al. (2022)
	<i>Rheum</i>	rhizome, leave	Izhaki (2002)*; Godard et al. (2009); Narayanan and Jadhav (2015a)
	<i>Rumex</i>	root, seed, leave, stem, flower	Izhaki (2002)*; Kengne et al. (2021)
<i>Rhamnaceae</i>	<i>Frangula</i>	bark	Godard et al. (2009);

Family	Genera	Plant parts	References
			Brkanac et al. (2015)
	<i>Rhamnus</i>	whole plant	Izhaki (2002)*; Godard et al. (2009)
	<i>Ventilago</i>	root, stem, bark	Izhaki (2002)*
<i>Rosaceae</i>	<i>Fragaria</i>	stem, leaves	Izhaki (2002)*
	<i>Prunus</i>	stem, bark	Izhaki (2002)*
<i>Saxifragaceae</i>	<i>Bergenia</i>	root, rhizome	Izhaki (2002)*; Rolta et al. (2022)
<i>Simaroubaceae</i>	<i>Bruceae</i>	rhizome	Izhaki (2002)*
	<i>Picramnia</i>	rhizome, stem, bark	Balderrama et al., (2001); Izhaki (2002)*
<i>Solanaceae</i>	<i>Lycium</i>	root bark	Yao et al. (2011)
<i>Vitaceae</i>	<i>Vitis</i>	leaves	Izhaki (2002)*
*See references within.			

In plants, emodin has an important role in the protection against some biotic and abiotic threats (Izhaki, 2002). For instance, its' presence in vegetable organs, especially in association with other anthraquinones, discourage herbivore to consume those plants (Nowak-Perlak et al., 2022). Therefore, to protect crops from certain herbivores, some recommend sicklepod (*Senna obtusifolia* L.) extracts as repellent, due to its high content in chrysophanol and emodin (Yue et al., 2018). Certain studies mention anthraquinones, such as emodin, rhein, chrysophanol and aloe-emodin, to have laxative activity (Van den Gorkom et al., 1999; Zheng et al., 2014). Fruit expresses higher amounts of emodin in the first stage of ripening, while its level decreases in ripe fruits. These aspects influence frugivorous birds to avoid consumption of immature fruits (Levey et al., 2007). Seed germination is also deterrent in unripe fruits due to the high level of emodin in the pulp (Tsahar et al., 2002). In this stage, the high amount of emodin is believed to protect seeds from insects and pathogens attack (Cipollini, 2000).

Emodin extracts can be used in plant protection against several phytopathogens, such as *Aspergillus niger*, *Botryosphaeria ribis*, *B. rhodina*, *Botrytis cinerea*, *Lasiodiplodia theobromae*, *Plasmopara viticola* and *Rhizoctonia solani* (Godard et al., 2009; Liu et al., 2009; de Barros et al. 2011). As well, it can inhibit foodborne contaminants such as *Bacillus cereus*, *Enterococcus faecalis*, *Escherichia coli* and *Staphylococcus aureus* which are pathogenic to humans (Liu et al., 2009). Moreover, emodin can protect plants also against some stressful abiotic factors, such as high light intensities (Sharifi-Rad et al., 2022).

FACTORS MODULATING EMODIN LEVELS IN PLANTS

The synthesis and accumulation of biologically active compounds in plants can be influenced by a series of abiotic factors. Seasonal variations, and environmental factors such as light intensity, temperature, macro- and microelements availability in the soil, as well as some osmotic stresses, like drought and salinity, can influence the level of plants secondary metabolites (Waterman and Mole, 1989). Therefore, to understand emodin role in plants it is important to explore how its synthesis is influenced by abiotic factors.

Seasons and light intensity are considered to influence emodin level in plants. In *Rheum undulatum* leaves, highest amount of emodin was detected in spring, and decreased during summer (Paneitz and Westendorf, 1999). Nutrient availability also influences emodin levels in leaves, as plants are redirecting their resources to support flowering and fruiting, by changing their metabolic activities (Izhaki, 2002).

In *Rhamnus purshiana* suspension cultures positive influence of photoperiod up to 12 h/day (Van den Berg et al., 1988) revealed a positive influence in increasing emodin content (Waterman and Mole, 1989). Plants organs of the same species exposed to increased light intensity also revealed higher emodin content (Waterman and Mole, 1989).

EXTRACTION METHODS OF EMODIN FROM *POLYGONACEAE* PLANTS

Conventional or modern extraction methods can be used for bioactive-compounds extraction from various plant parts and species. However, conventional extraction methods demand high amounts of solvents and reactants, which generate chemical wastes. Moreover, samples are highly exposed to thermal degradation during extraction and purification (Beňová *et al.*, 2010). Degradation studies revealed that 8 hours of dry heat exposure, at 105°C, lead to 82.05% emodin amount recovery (Narayanan *et al.*, 2015b).

Emodin extraction from *Polygonaceae* plants is possible through various methods, such as classical methods, like decoction, maceration, percolation, stirring extraction, heat-reflux, or Soxhlet extraction, as well as non-conventional methods like ultra-sonication, sublimation, or supercritical carbon dioxide (CO₂) fluid extraction.

Decoction is an empiric extraction method, for which the vegetal material is boiled for up to 1h in water at 95-98°C. The resulting mixture is filtered and the solid phase can be re-extracted in the same way until exhausting. The Bonträger's reaction can be used to reveal whether the sediment has become exhausted (Khoomsab and Khoomsab, 2019). At this point, all decoction extracts should be combined, filtered again and evaporated to dryness, on a boiling water bath, in order to yield the crude extracts (Sakulpanich and Gritsanapan, 2008).

Similar to decoction there is **the boiling method**. For this the vegetal material is boiled in water, in 1:100 (w/v) ratio, for several hours (e.g. 7h). Then, the filtered extract is evaporated, as previously mentioned, in order to yield the crude extract (Khoomsab and Khoomsab, 2019).

Maceration can be performed in different solvents, but most commonly in ethanol. The extraction is carried out at room temperature. As in the previous methods, this extraction type can also be repeated, until exhaustion. If so, the maceration extracts of the same sample should be combined, filtered and evaporated, in order to yield the crude extract (Sakulpanich and Gritsanapan, 2008).

Percolation is another classical extraction method performed at room temperature. It can be performed using different solvents, and solvent concentrations. For example, Sakulpanich and Gritsanapan (2008) mention 70% ethanol as solvent for anthraquinone extraction.

The **heat-reflux extraction method** is sometime referred as reflux assisted extraction. For this method the sample is put together with the solvent (e.g. ethanol), in a round bottom flask, and refluxed for different time intervals on water bath. After extraction, the supernatant is collected by centrifugation of repeat filtration and crude extract can be yielded by evaporation.

Among the classical extraction methods, the Soxhlet is considered the most reliable. Moreover, the **Soxhlet extraction** is also the most popular technique. It is used to extract those compounds with limited solubility in a certain solvent, in which most of impurities are insoluble. This solvent extraction from solid samples, is commonly known as solid-liquid extraction, also referred to as lixiviation. It is a simple and effective method, routinely applied, and sometime used as a standard technique to compare the performance of modern extractions methods (Zygler *et al.*, 2012). For this technique a Soxhlet equipment is used. At first, the sample is placed in a paper thimble-holder positioned above a solvent containing flask, which is placed in the Soxhlet apparatus. The solvent flask is gently heated through a mantle, in a controlled manner. During extraction, vapors of the heated solvent reach the condenser and the condensing solvent drips into the thimble percolating the sample. Thus, the extraction chamber is gradually filled with condensed fresh solvent which slowly extract the soluble compounds from the sample. When the liquid reaches an overflow level, a siphon aspirates the liquid phase containing the extracted compounds and unloads it into the distillation flask (figure 2). The solvent is then recirculated through the system to repeat the extraction until matrix exhausting. The process is allowed to continue for as long as necessary to carry as much of the extracted analytes in the bulk liquid (Luque de Castro and García Ayuso, 2000).

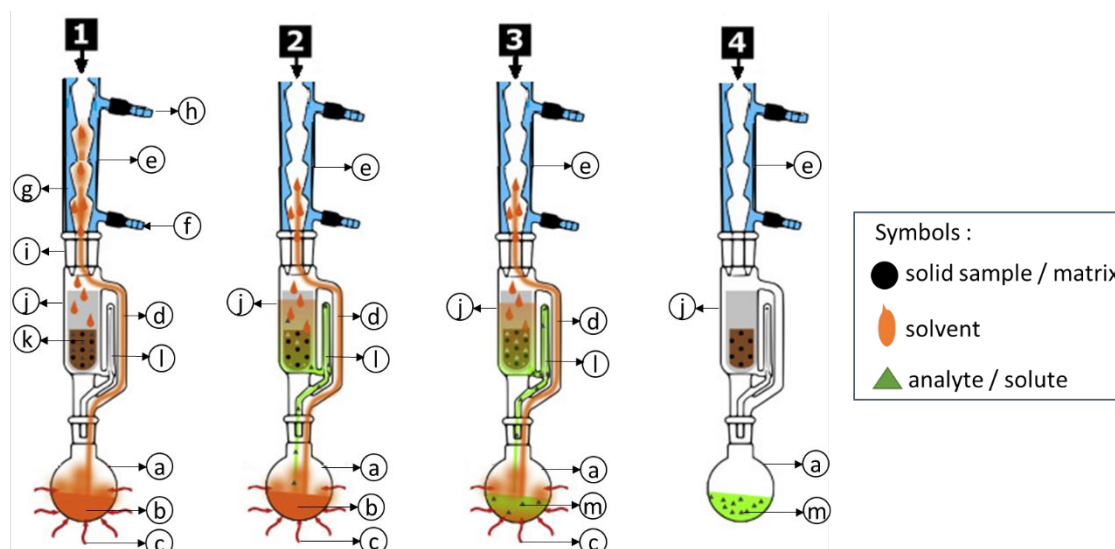


Fig. 2 - The workflow extraction in Soxhlet apparatus (Weggler et al., 2020)

Legend: 1) Solid matrix is placed in a thimble-holder lined with filter paper. The solvent containing flask is heated. The condensed solvent starts to percolate the sample 2) Solutes are transferred from the extraction chamber, through the siphon, into the lower reservoir. From the distillation chamber the solvent is recirculated through the sample to continue the extraction 3) The extraction is maintained in continuous repetition to carry as much of the extracted analytes in the bulk liquid. 4) Exhaustive extraction is complete. a. Round bottom flask / boiling flask / distillation flask / lower reservoir, b. extracting solvent, c. heating mantle, d. distillation path / vaporized solvent flux, e. water cooled condenser, f. cold water supply nozzle, g. cooling water, h. cold water outlet nozzle, i. expansion adapter, j. extractor containing a paper-thimble in which the solid sample is placed / extraction chamber/ upper reservoir, k. chopped vegetal sample, l. siphon, m. analytes in the bulk liquid.

Emodin extracted from *Rumex acetosa* and *Polygonum cuspidatum* roots using the Soxhlet method, revealed high anti-inflammatory and antioxidant properties (Souto, 2011).

Through Soxhlet, extraction can be carried out with different solvents, such as ethanol, methanol, acetone, ethyl acetate, dichloromethane or chloroform (Arvindekar et al., 2015). However, these organic solvents are having some drawbacks, including various levels of toxicity, especially if the solvent traces remain present in the final product, not to mention the disposal problems. As an alternative, natural deep eutectic solvents can be used to extract anthraquinones. The extraction method appropriate when using these solvents is the **ultrasound-assisted extraction**. Among 13 combinations of natural deep eutectic solvents used for anthraquinones ultrasound extraction from *Rheum palmatum* L. best results were obtained with lactic acid/glucose ratio of 5:1 (mol/mol), and 10% of water (v/v) (Wu et al., 2018).

To extract 1,8-dihydroxyanthraquinones, such as aloe-emodin, chrysophanol, emodin, ehysson and rhein, ultra-sonication assisted extraction is also possible in organic solvents. The extraction can be made at a sample: solvent ratio of 1:20 w/v in ethanol. Best extraction conditions were obtained when samples were sonicated at 40 Hz frequency, 100 W, for 45 min. Longer time exposure did not improved anthraquinone extraction through this method, either due to matrix exhausting, solvent saturation, or releasement of other components from the plant matrix that interfere with further extraction. Separation of the ultra-sonicated extract from the solid phase can be made by centrifugation (Arvindekar et al., 2015).

High purity anthraquinones (emodin, chrysophanol, and physcion), were successfully isolated from whole plant of *Rumex japonicus*, by two times repeated ultrasonication in dichloromethane for 30 min at room temperature (Guo et al., 2011). Anthraquinones from *Rheum emodi* (rhein, aloe emodin, emodin, chrysophanol and physcion) with important pharmacological properties were extracted through ultra-sonication method (Arvindekar et al., 2015).

Sublimation assisted extraction is also suitable for these compounds. Within this method the sample is mixed with the solvent (e.g. ethanol) in a 1:50 (w/v) ratio. For sublimation the sample flask was strongly heated for up to 10 minutes, until no yellow fumes arise from the sample. After cooling off the equipment at room temperature, the sublimed extracted anthraquinones, adhered to the condenser walls, can be dissolved in ethanol (Arvindekar et al., 2015).

The supercritical fluid extraction uses a high-pressure CO₂ pump for carbon dioxide displacement. Sample is to be put in an extraction vessel on a thermostatic bath. The CO₂ - solvent mixture is pumped in the extraction vessel using a HPLC pump. For anthraquinone supercritical fluid extraction, there were tested various parameters range. Before performing the extraction, it is mandatory to stabilize the extraction parameters of the system, and maintain them for 30 minutes. If the extraction is performed at 60°C temperature, 20MPa pressure, for 180 minutes, with a mixture of CO₂ and 15% ethanol (v/v) as cosolvent, the best flow rate to obtain higher extraction yields is 1mL/min, from a tested range of 1 to 4mL/min. From a range of 15 to 25MPa pressure and 40 to 60°C temperature, best solubility and yield is seen at 20 MPa and 60°C, with 1mL/min flow rate and 15% ethanol (v/v) as cosolvent. Moreover, augmented ethanol addition to the supercritical CO₂, also increases the extraction yield. From a range of 5 to 35% ethanol (v/v), into an extraction performed at 40°C and 15MPa pressure, for 180 min extraction time, and flow rate of 1 mL/min, best yield results are obtained with 35% ethanol as cosolvent (Santos *et al.*, 2017). Comparing supercritical fluid extraction with sonication, Lu *et al.* (2006) obtained little lower yield of emodin and physcion from *Polygonum cuspidatum* Sied. et Zucc. with the first method, but of much higher purity and concentration of target compounds in the extract, compared to the sonicated extract.

PHARMACOLOGICAL PROPERTIES OF EMODIN

It is well known that emodin has a lot of pharmacological valuable properties. Years of research confirm that emodin, apart from the classical laxative action, has anticancer, anti-inflammatory, antioxidant, antibacterial, antiviral, and anti-diabetes activity (Dong *et al.*, 2016). In alternative medicine, emodin is used for preventing various health problems. Several studies mention emodin to have protective effect on the epithelium, cardiovascular system, eyes, kidney, pancreas, oral cavity, lungs, nervous system, liver and skeleton (references within Arvindekar *et al.* 2015). This protective effects of emodin are presented in the following figure (figure 3).



Fig. 3 - Emodin protective effects on human body (Cui *et al.*, 2020)

Due to the current interest on alternative therapies, the use of emodin as vegetal origin "medicine" should be considered for preventive and therapeutic treatments. However, it should be taken into account that the long-term use of this biologically active compound can cause hepatotoxicity or renal toxicity, these being influenced by the concentration and the administered dose.

CONCLUSIONS

Continuous demand for therapeutic natural-originated bioactive compounds encourages the research for convenient extraction methods. Beside classical extraction methods, like decoction, maceration, percolation, stirring extraction, heat-reflux, or Soxhlet extraction, non-conventional methods like ultra-sonication, sublimation, or supercritical carbon dioxide (CO₂) fluid extraction were adapted for emodin extraction. Due to some drawbacks of the organic solvents, few studies have showed that emodin can also be extracted using certain combinations of natural deep eutectic solvents, but only in association with specific extraction methods.

The need for less expensive, environmentally friendly and more efficient extraction-methods derives from emodin wide beneficial properties. Emodin pharmacological benefits indicating a high potential to become a preventive and therapeutic drug for various diseases.

ACKNOWLEDGEMENT

We are grateful to Dr. Daniel Kazimir KURZELUK for his advices and kind help.

REFERENCES

- [1] Akkol, E. K., Tatlı, I. I., Karatoprak, G. S., A`gar, O. T., Yücel, Ç., Sobarzo-Sánchez, E., & Capasso, R. (2021). Is Emodin with anticancer effects completely innocent? Two sides of the coin. *Cancers*, 13, 2733. <https://doi.org/10.3390/cancers13112733>.
- [2] Arvindekar, A. U., Pereira, G. R., & Laddha, K. S. (2015). Assessment of conventional and novel extraction techniques on extraction efficiency of five anthraquinones from *Rheum emodi*. *Journal of food science and technology*, 52(10), 6574–6582. <https://doi.org/10.1007/s13197-015-1814-3>
- [3] Balderrama, L., Braca, A., Garcia, E., Melgarejo, M., Pizza, C., & De Tommasi, N. (2001). Triterpenes and anthraquinones from *Picramnia sellowii* Planchon in Hook (Simaroubaceae). *Biochemical Systematics and Ecology*, 29, 331–333.
- [4] Beňová, B., Adam, M., Pavlíková, P., & Fischer J. (2010). Supercritical fluid extraction of piceid, resveratrol and emodin from Japanese knotweed. *J. Supercrit. Fluids*, 51, 325–330, <https://doi.org/10.1016/j.supflu.2009.10.009>.
- [5] Boonnak, N. (2006). Chemical constituents from the barks of *Cratoxylum formosum* ssp. *pruniflorum*. Master of Science Thesis in Organic Chemistry, Prince of Songkla University, Songkhla, Thailand
- [6] Brkanac, S.R., Gerić, M., Gajski, G., Vujčić, V., Garaj-Vrhovac, V., Kremer, D., & Domijan, A.-M. (2015). Toxicity and antioxidant capacity of *Frangula alnus* Mill. bark and its active component emodin. *Regul. Toxicol. Pharmacol.*, 73, 923–929. <https://doi.org/10.1016/j.yrtph.2015.09.025>.
- [7] Cipollini, M. L. (2000). Secondary metabolites of vertebrate-dispersed fruits: evidence for adaptive functions. *Revista Chilena de Historia Natural*, 73, 421– 440.
- [8] Cui, Y., Chen, L. J., Huang, T., Ying, J. Q., & Li, J. (2020). The pharmacology, toxicology and therapeutic potential of anthraquinone derivative emodin. *Chinese Journal of Natural Medicines*, 18(6), 425–435. [https://doi.org/10.1016/S1875-5364\(20\)30050-9](https://doi.org/10.1016/S1875-5364(20)30050-9)
- [9] de Barros, I. B., de Souza-Daniel, J. F., Pinto, J. P., Rezendo, M. I., Filho, R. B. & Ferreira, D. T. (2011). Phytochemical and antifungal activity of anthraquinones and root and leaf extracts of *Coccoloba mollis* on phytopathogens. *Braz. Arch. Biol. Technol.*, 54, 535–541.
- [10] Ee, G. C. L., & Ng, K. N. (2004). Larvicidal anthraquinones and triterpenes from *Ploiarium alternifolium* (Theaceae). *Asian Journal of Chemistry*, 16, 429-433.

- [11] Ghosh, D., Mitra, P., Ghosh, T., Gupta, S., Basu, B., & Mitra, P.K., 2014. Isolation of emodin from the leaves of *Amaranthus spinosus* L. (*Amaranthaceae*). *World Journal of Pharmaceutical Research*, 3(6): 1790-1800.
- [12] Godard, S., Slacanin, I., Viret, O., & Gindro, K. (2009). Induction of defence mechanisms in grapevine leaves by emodin- and anthraquinone-rich plant extracts and their conferred resistance to downy mildew. *Plant Physiology and Biochemistry*, 47, 827–837. <https://doi.org/10.1016/j.plaphy.2009.04.003>.
- [13] Guo, S., Feng, B., Zhu, R., Ma, J., & Wang, W., (2011). Preparative isolation of three anthraquinones from *Rumex japonicus* by high-speed counter-current chromatography. *Molecules*, 16(2), 1201-1210. <https://doi.org/10.3390/molecules16021201>.
- [14] Ismaiel, A. A., Rabie G. H., & Abd El-Aal, M. A. (2016). Antimicrobial and morphogenic effects of emodin produced by *Aspergillus awamori* WAIR120. *Biologia*, 71(5), 464—474. <https://doi.org/10.1515/biolog-2016-0067>
- [15] Izhaki, I., (2002). Emodin – a secondary metabolite with multiple ecological functions in higher plants. *New Phytologist*, 155: 205-217. <https://doi.org/10.1046/j.1469-8137.2002.00459.x>.
- [16] Kengne, I. C., Feugap, L. D. T., Njouendou, A. J., Ngnokam, C. D. J., Djamalladine, M. D., Ngnokam, C. D. J., Djamalladine, M. D., Ngnokam, D., Voutquenne-Nazabadioko, L., & Tamokou, J. D. D. (2021). Antibacterial, antifungal and antioxidant activities of whole plant chemical constituents of *Rumex abyssinicus*. *BMC Complement. Med. Ther.*, 21, 164. <https://doi.org/10.1186/s12906-021-03325-y>.
- [17] Khoomsab, R., & Khoomsab, K. (2019). Extraction and determination of anthraquinone from herbal plant as bird repellent. *Science & Technology Asia*, 24 (1), 14-20. <https://doi.org/10.14456/scitechasia.2019.2>
- [18] Levey, D., Tewksbury, J.J., Izhaki, I., Tsahar, E., & Haak, D. (2007). Evolutionary ecology of secondary compounds in ripe fruit: Case studies with capsaicin and emodin. Chapter 2 In Dennis A.J. et al. (eds.). *CAB International. Seed dispersal*. <https://doi.org/10.1079/9781845931650.0037>.
- [19] Li, X., Shan, C., Wu, Z., Yu, H., Yang, A., & Tan, B., (2020). Correction to emodin alleviated pulmonary inflammation in rats with LPS-induced acute lung injury through inhibiting the mTOR/HIF-1a/VEGF signaling pathway. *Inflammation Res.*, 69 (7), 711. <https://doi.org/10.1007/s00011-020-01344-y>.
- [20] Lin, C.W., Tsai, F.J., Tsai, C.H., Lai, C.C., Wan, L., Ho, T.Y., & Chao, P.D.L. (2005). Anti- SARS coronavirus 3C-like protease effects of *Isatis indigotica* root and plant-derived phenolic compounds. *Antiviral Research*, 68 (1), 36–42.
- [21] Lin, S., Wang, X., Tang, R. W., Lee, H. C., Chan, H. H., Choi, S., Dong, T. T., Leung, K. W., Webb, S. E., Miller, A. L., & Tsim, K. W. (2022). The Extracts of *Polygonum cuspidatum* Root and Rhizome Block the Entry of SARS-CoV-2 Wild-Type and Omicron Pseudotyped Viruses via Inhibition of the S-Protein and 3CL Protease. *Molecules (Basel, Switzerland)*, 27(12), 3806. <https://doi.org/10.3390/molecules27123806>
- [22] Liu, S.-Y., Lo, C.-T., Marthandam Asokan, S., Leu, Y.-L., Jen, B.-Y., Peng, K.-C. (2009). Study on the Anthraquinones separated from the cultivation of *Trichoderma harzianum* strain Th-R16 and their biological activity. *Journal of Agricultural and Food Chemistry*, 57, 7288-7292. <https://doi.org/10.1021/jf901405c>.
- [23] Lu, H. M., Ni, W. D., Liang, Y. Z., & Man, R. L., (2006). Supercritical CO₂ extraction of emodin and physcion from *Polygonum cuspidatum* and subsequent isolation by semipreparative chromatography. *J Sep Sci.*, 29(14), 2136-2142. <https://doi.org/10.1002/jssc.200600012>.
- [24] Lu, P., Zhao, X., Cui, T. (2010). Production of emodin from *Aspergillus ochraceus* at preparative scale. *Afr. J. Biotechnol.*, 9, 512–517.
- [25] Luque de Castro, M. D., & García Ayuso, L. E. (2000). Environmental applications - Soxhlet Extraction. In Ian D. Wilson (ed.), *Encyclopedia of Separation Science*, Academic Press, pp 2701-2709, <https://doi.org/10.1016/B0-12-226770-2/06681-3>.

- [26] Kikuchi, N., Teshiba, M., Tsutsumi, T., Fudou, R., Nagasawa, H., & Sakuda, S. (2011). Endocrocin and its derivatives from the Japanese mealybug *Planococcus kraunhiae*. Biosci. Biotechnol. Biochem., 75 (4), 764–767.
- [27] Narayanan, S., & Jadhav, A. P. (2015a). Simultaneous Estimation of Aloe Emodin and Emodin from Rheum emodi, Cassia alata and Aloes by HPTLC. Indian journal of pharmaceutical sciences, 77(6), 783–787. <https://doi.org/10.4103/0250-474x.174994>
- [28] Narayanan, S., Jadhav, A. P., & Kadam, V. J. (2015b). Forced Degradation Studies of Aloe Emodin and Emodin by HPTLC. Indian journal of pharmaceutical sciences, 77(6), 795–798. <https://doi.org/10.4103/0250-474x.174997>
- [29] Nowak-Perlak, M., Bromke, M.A., Ziółkowski, P. & Wozniak, M. (2022). The Comparison of the Efficiency of Emodin and Aloe-Emodin in Photodynamic Therapy. Int. J. Mol. Sci., 23, 6276. <https://doi.org/10.3390/ijms23116276>
- [30] Paneitz, A., & Westendorf, J., (1999). Anthranoid contents of rhubarb (*Rheum undulatum* L.) and other *Rheum* species and their toxicological relevance. European Food Research and Technology, 210, 97–101.
- [31] Qiu, X., Gong, L., Xin, X., An, F. (2021). Enhancement of Emodin Production by Medium Optimization and KH₂PO₄ Supplementation in Submerged Fermentation of Marine-Derived *Aspergillus favipes* HN4-13. Mar. Drugs, 19, 421. <https://doi.org/10.3390/md19080421>
- [32] Rehorska, R., Lernbeiss, J., Plinter, M., Müller, M., & Pfeifhofer, H.V. (2013). Simultaneous Detection of (+)-Catechin, Juglone, Emodin, Piceatannol, Physcion and Resveratrol by TLC in a Two Step Procedure - a "Quick and Dirty" Approach. Phyton (Horn, Austria), 53 (1), 125-132.
- [33] Rolta, R., Goyal, M., Sharma, S., Bharaj, D., Salaria, D., Upadhyay, N.K., Lal, U.R., Dev, K., & Sourirajan, A. (2022). Bioassay Guided Fractionation of Phytocompounds from *Bergenia ligulata*: A synergistic approach to treat drug resistant bacterial and fungal pathogens. Pharmacological Research - Modern Chinese Medicine, 3, 100076, <https://doi.org/10.1016/j.prmcm.2022.100076>.
- [34] Sakulpanich, A., & Gritsanapan, W. (2008). Extraction method for high content of anthraquinones from Cassia fistula pods. Journal of Health Research, 22(4), 167-172.
- [35] Santos, Ê. R. M., Oliveira, H. N. M., Oliveira, E. J., Azevedo, S. H. G., Anderson J.A., Medeiros, A. M., Dariva, C., & Sousa, E. M. B. D., (2017). Supercritical fluid extraction of *Rumex acetosa* L. roots: yield, composition, kinetics, bioactive evaluation and comparison with conventional techniques. The Journal of Supercritical Fluids, 122, 1-9.
- [36] Sharifi-Rad, J., Herrera-Bravo, J., Kamiloglu, S., Petroni, K., Mishra, A.P., Monserrat-Mesquida, M., Sureda, A., Martorell, M., Aidarbekovna, D.S., Yessimsiitova, Z., Ydyrys, A., Hano, C., Calina, D., & Cho, W.C. (2022). Recent advances in the therapeutic potential of emodin for human health, Biomedicine & Pharmacotherapy, 154, 113555
- [37] Souto, A.A., (2011). Process of obtainment of trans-resveratrol and/or emodin and nutraceutical compositions containing them, US7977516 B2, 2011. <https://www.google.ch/patents/US7977516>.
- [38] Tsahar, E., Friedman, J. & Izhaki, I. (2002). Impact on fruit removal and seed predation of a secondary metabolite, emodin, in *Rhamnus alaternus* fruit pulp. Oikos, 99, 290– 299.
- [39] Van den Berg, A. J. J., Radema, M. H., & Labadie, R. P. (1988). Effects of light on anthraquinone production in *Rhamnus purshiana* suspension cultures. Phytochemistry, 27(2), 415-417. [https://doi.org/10.1016/0031-9422\(88\)83110-8](https://doi.org/10.1016/0031-9422(88)83110-8).
- [40] Van den Gorkom B. A. P., De Vries E. G. E., Karrenbeld, A., & Kleibeuker, J. H. (1999). Anthranoid laxatives and their potential carcinogenic effects. Alimentary Pharmacology and Therapeutics, 13, 443– 452.
- [41] Waterman, P.G., & Mole, S., (1989). Extrinsic factors influencing production of secondary metabolites in plants. In: Bernays EA, ed. Insect–plant interactions, Vol. 1. Boca Raton, FL, USA: CRC Press, 107–134.
- [42] Weggler, B.A., Gruber, B., Teehan, P., Jaramillo, R., & Dorman, F.L. (2020). Chapter 5 - Inlets and sampling. In Nicholas H. Snow (ed.), Separation Science and Technology, Academic Press, Volume 12, pp. 141-203, <https://doi.org/10.1016/B978-0-12-813745-1.00005-2>.

- [43] Wu, Y. C., Wu, P., Li, Y. B., Liu, T. C., Zhang, L., & Zhou, Y. H. (2018). Natural deep eutectic solvents as new green solvents to extract anthraquinones from *Rheum palmatum* L. Royal Society of Chemistry Advances, 8, 15069. <https://doi.org/10.1039/c7ra13581e>.
- [44] Xiao, H.H., Zhang, Y., Cooper, R., Yao, X.S., & Wong, M.S. 2016. Phytochemicals and potential health effects of *Sambucus williamsii* Hance (Jiegumu). Chinese Medicine, 11, 36.
- [45] Yao, X., Peng, Y., Xu, L.J., Li, L., Wu, Q.L., & Xiao P.G. (2011). Phytochemical and biological studies of *Lycium medicinal plants*, Chem. Biodiver. 8, 976–1010.
- [46] Yue, Z. , Tseng, T. & Lashley, M. (2018) Characterization and Deer-Repellent Property of Chrysophanol and Emodin from Sicklepod Weed. American Journal of Plant Sciences, 9, 266-280. , <https://doi.org/10.4236/ajps.2018.92022>.
- [47] Zhang, Z., Dai, L., Wang, H., Chang, X., Ren, S., Lai, H., & Liu L., (2021). Phytochemical profiles and antioxidant, anticholinergic, and antidiabetic activities of *Odontites serotina* (Lam.) Dum. European Journal of Integrative Medicine, 44, 101340. , <https://doi.org/10.1016/j.eujim.2021.101340>
- [48] Zheng, Y. F., Liu, C. F., Lai, W. F., Xiang, Q., Li, Z. F., Wang, H., & Lin, N. (2014). The laxative effect of emodin is attributable to increased aquaporin 3 expression in the colon of mice and HT-29 cells. Fitoterapia, 96, 25–32. <https://doi.org/10.1016/j.fitote.2014.04.002>
- [49] Zygler, A., Słomińska, M., & Namieśnik, J. (2012). Soxhlet extraction and new developments such as soxtec, in comprehensive sampling and sample preparation, Janusz Pawliszyn (ed.), Academic Press, pp. 65-82, <https://doi.org/10.1016/B978-0-12-381373-2.00037-5>.